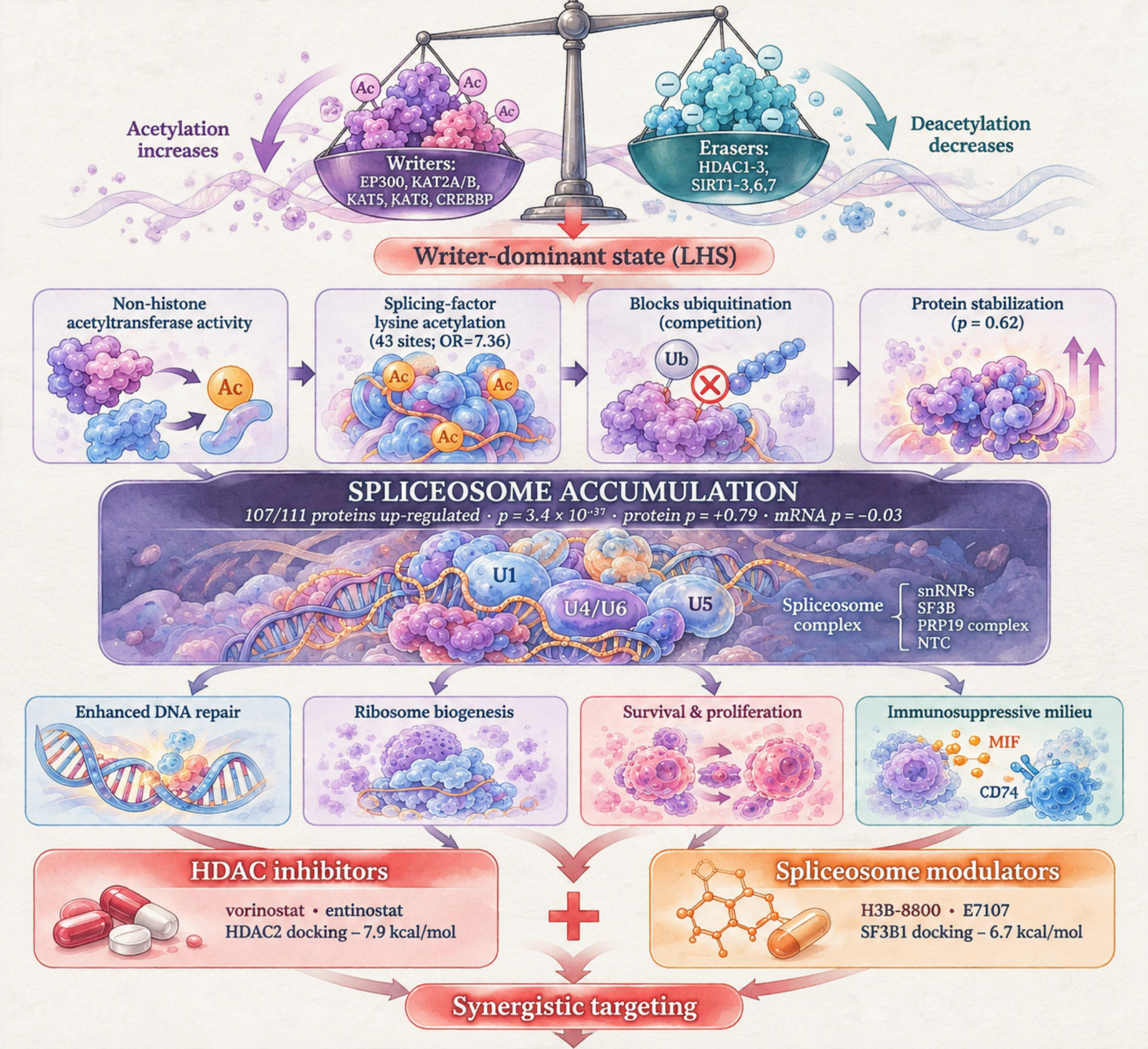

Acetylation increases
Writers:
EP300, KAT2A/B,
KAT5, KAT8, CREBBP
Erasers:
HDAC1-3,
SIRT1-3,6,7
Deacetylation decreases
Writer-dominant state (LHS)
Non-histone acetyltransferase activity
Splicing-factor lysine acetylation (43 sites; OR=7.36)
Blocks ubiquitination (competition)
Ub
Protein stabilization (p = 0.62)
SPLICEOSOME ACCUMULATION
107/111 proteins up-regulated · p = 3.4 x 10⁻³⁷ · protein p = +0.79 · mRNA p = −0.03
U1
U4/U6
U5
Spliceosome complex
snRNPs
SF3B
PRP19 complex
NTC
Enhanced DNA repair
Ribosome biogenesis
Survival & proliferation
Immunosuppressive milieu
MIF
CD74
HDAC inhibitors
vorinostat • entinostat
HDAC2 docking – 7.9 kcal/mol
Spliceosome modulators
H3B-8800 • E7107
SF3B1 docking – 6.7 kcal/mol
Synergistic targeting